KG COLLEGE OF ARTS AND SCIENCE
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KGiSL Campus, Saravanampatti, Coimbatore – 641 035

JAVA PROJECT REPORT ON

INSURANCE POLICY MANAGEMENT AND PREMIUM REMINDER SYSTEM

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DECLARATION

I hereby declare that the project entitled "INSURANCE POLICY MANAGEMENT AND PREMIUM REMINDER SYSTEM**"** submitted in partial fulfillment of the requirements for the course java programming is a **record of my original work** carried out under the guidance of Mr.Aravind, Technical Trainer, Technical Department, KG College of Arts and Science.

I further declare that this project has not been submitted previously, either in part or in full, for the award of any degree, diploma, or other similar title, at this or any other institution.

I have acknowledged all sources of information used in this project wherever applicable.

April 2026

Coimbatore

HARINI S

(your Name)

CERTIFICATE

This is to certify that the project entitled **"INSURANCE POLICY MANAGEMENT AND PREMIUM REMINDER SYSTEM"** is a **bona fide work** carried out by HARINI S , I-B.SC COMPUTER SCIENCE , **KG College of Arts & Science**, in partial fulfillment of the requirements for the course java programming.

This project has been completed under my supervision and guidance during the academic period **April 2026**.

Place: COIMBATORE

Date:10/04/2026

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Guide: **Head of the Department**(Technical Trainer, KGCAS) (Technical Head, KGCAS)

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ABSTRACT

The Insurance Policy Management and Premium Reminder System is designed to simplify the process of managing insurance policies and ensuring timely premium payments. This system helps users store and organize policy details such as policy ID, holder name, premium amount, and due dates in a structured manner. It reduces manual effort and minimizes the risk of missing premium deadlines.The application provides features to add, view, update, and delete policy records efficiently. One of the key functionalities of the system is the automated reminder mechanism, which alerts users about upcoming premium due dates. This improves financial planning and avoids penalties due to late payments.The system is implemented using basic programming concepts such as classes, objects, and data structures, making it easy to understand and maintain. Overall, this project enhances accuracy, saves time, and provides a user-friendly approach to managing insurance policies.An Insurance Policy Management and Premium Reminder System is designed to simplify and automate the process of managing insurance policies and tracking premium payments. This system helps users store and organize policy details such as policy ID, policyholder name, premium amount, and due dates in a structured manner. It ensures easy access, modification, and monitoring of policy information, reducing manual effort and minimizing errors. Additionally, the system includes a reminder feature that alerts users about

upcoming premium due dates, helping them avoid penalties and policy lapses. By using this system, both insurance companies and customers can improve efficiency, accuracy, and overall policy management.An extra advantage of this system is its ability to enhance user convenience and reliability through automation. The reminder functionality can be extended to send notifications via SMS, email, or app alerts, ensuring users never miss important deadlines. It also supports better financial planning by providing clear insights into upcoming payments and policy status. Furthermore, the system can be upgraded with features like report generation, data analytics, and secure data storage, making it more scalable and suitable for real-world applications in the insurance industry.

1. INTRODUCTION

The Insurance Policy Management and Premium Reminder System is a software application designed to efficiently manage insurance policy details and ensure timely premium payments. In today's fast-paced world, individuals often forget important payment deadlines, which can lead to policy lapses and financial loss. This system aims to solve such problems by maintaining accurate records of policyholders, their policies, premium amounts, and due dates in a structured manner.

The system provides a user-friendly interface to add, update, delete, and view policy details. One of its key features is the premium reminder functionality, which alerts users before the due date, helping them make payments on time. This reduces the risk of penalties and ensures continuous policy coverage.

By automating manual tasks and organizing data efficiently, the system improves accuracy, saves time, and enhances overall management of insurance policies. It can be implemented using programming languages like Java and can be further extended with features such as email or SMS notifications, database integration, and user authentication for better security.

Overall, this system serves as a reliable tool for both individuals and organizations to manage insurance policies effectively and avoid missed premium payments.

1.1 Objective

1.Centralized Policy Management

To store and manage all insurance policy details (policy ID, holder name, premium amount, due date) in one system.

2.Automated Premium Reminders

To notify users about upcoming premium due dates and avoid late payments.

3.Easy Data Access and Update

To allow users to quickly add, view, update, or delete policy records.

4.Reduce Manual Work

To minimize paperwork and human errors by automating the management process.

5.Improved Customer Experience

To ensure users never miss payments and maintain continuous policy coverage.

6.Data Organization and Security

To maintain structured and secure storage of policy information.

7.Time Efficiency

To save time by providing quick retrieval of policy details and reminders.

8.Scalability

To design a system that can handle multiple users and policies efficiently.

1.2 Overview

The Insurance Policy Management and Premium Reminder System is a software application designed to help users and insurance companies efficiently manage insurance policies and ensure timely premium payments. This system maintains detailed records of policyholders, including policy ID, customer name, premium amount, and due dates.

The main purpose of the system is to simplify the process of tracking multiple insurance policies and to reduce the risk of missing premium payments. It provides automated reminders to users before the due date, helping them avoid penalties or policy lapses.

This system also allows administrators to add, update, delete, and view policy details easily. By digitizing policy management, it reduces manual paperwork, improves accuracy, and saves time.
Overall, the system enhances convenience, reliability, and efficiency in handling insurance-related activities, making it beneficial for both customers and insurance providers.
1.3 features

1. Policy Management

Add, update, delete, and view insurance policies

Store details like policy ID, holder name, premium amount, due date

Categorize policies (Life, Health, Vehicle, etc.)

2. Customer Management

Maintain customer details (name, contact, email, address)

Link multiple policies to a single customer

Easy retrieval of customer history

3. Premium Reminder System

Automatic reminders before due date

Notifications via SMS / Email (or system alerts)

Custom reminder settings (e.g., 7 days before, 1 day before)

4. Payment Tracking

Record premium payments

Track paid, pending, and overdue premiums

Maintain payment history

1. PROGRAM ANALYSIS

The Insurance Policy Management and Premium Reminder System is designed to manage policyholder details, track premium payments, and send reminders before due dates. It reduces manual work and helps users avoid missing payments.

2.1 Problem Definition

In today's fast-paced world, many individuals and organizations struggle to manually manage their insurance policies and keep track of premium due dates. Traditional methods such as paper records, spreadsheets, or memory-based tracking often lead to missed premium payments, policy lapses, and financial losses.

There is no centralized system that efficiently organizes policy details such as policy ID, holder information, premium amount, and due dates while also providing timely reminders. As the number of policies increases, managing them manually becomes more complex, error-prone, and time-consuming.

Additionally, the absence of automated notifications results in users forgetting important deadlines, which may lead to penalties or termination of insurance coverage. This creates inconvenience and reduces overall trust in policy management processes.

Therefore, there is a need for a reliable and automated system that can store, manage, and monitor insurance policies while sending timely premium reminders to users, ensuring better organization, reduced errors, and improved user experience.

2.2 Project Overview

The Insurance Policy Management and Premium Reminder System is a software application designed to efficiently manage insurance policies and ensure timely premium payments. This system helps both insurance companies and policyholders by automating policy tracking, premium calculations, and reminder notifications.

The main purpose of this project is to eliminate manual errors and reduce the chances of missed premium payments. It stores essential details such as policy ID, policyholder name, premium amount, due date, and policy status in a structured format. Users can easily add, update, delete, and view policy information through the system.

2.3 Scopes and Limitations

Scope

The scope defines what your system can do and where it can be used:

1. Policy Management

Add, update, delete insurance policies

Store customer details (name, policy ID, premium, due date)

2. Premium Tracking

Track premium payment status

Maintain records of paid and unpaid premiums

3. Reminder System

Send reminders before due date

Alerts for overdue payments

4. User-Friendly Interface

Simple interface for users/admin

Easy navigation and data access

5. Data Storage

Store all policy details in database or file system

Retrieve data quickly when needed

6. Report Generation

Generate reports for:

Due premiums

Paid premiums

Customer details

7. Multi-user Support (Future Scope)

Can be extended for multiple users (admin + customers)

These are the restrictions or drawbacks:

1. Limited Automation

Basic system may not support automatic SMS/email reminders

Needs manual checking in simple Java programs

2. No Real-Time Integration

Not connected to real insurance companies or banks

Payments may not be updated automatically

3. Security Issues

Basic system may lack strong security features

Risk of data loss or unauthorized access

4. Scalability Problem

Difficult to handle large number of users/policies

Works best for small-scale applications

5. Manual Data Entry

User must enter all details manually

Human errors may occur

6. Limited Platform

If developed in core Java, may work only on desktop

No mobile/web access unless upgraded

7. No Backup System

Data may be lost if system crashes (if backup not implemented

1. SYSTEM DESIGN

The Insurance Policy Management and Premium Reminder System is designed using a simple and efficient architecture that helps users manage their insurance policies and receive timely reminders for premium payments. The system consists of three main layers: the presentation layer, where users interact through a user interface; the application layer, which handles the core business logic such as adding, updating, and checking policies; and the data layer, where all policy details are securely stored.

3.1 Modules

The key modules for an Insurance Policy Management and Premium Reminder System: **1. Menu Display Module**
This module typically displays options such as Add Policy, View Policies, Update Policy, Delete Policy, and Set Premium Reminder. When the system starts, the menu is shown to the user, and based on their choice, the corresponding functionality is executed. It ensures smooth interaction between the user and the system by continuously prompting for input until the user chooses to exit. **2. Order Selection Module**
This module ensures that the correct operation is executed efficiently and prevents invalid choices by displaying error messages and prompting the user again.

For example, if the user selects:

1 → Add Policy, it calls the add policy function

2 → View Policy, it displays all stored policies

3 → Update Policy, it modifies existing policy details

4 → Delete Policy, it removes a policy

5 → Reminder, it shows or sets premium reminders

3. Bill Calculation Module
The Bill Calculation Module is responsible for calculating the total premium amount that a policyholder needs to pay. It ensures accurate billing based on policy details, due dates, and any additional charges.

Functions of this Module:

1.Premium Calculation

Calculates the base premium amount of the policy.

Based on policy type, duration, and coverage.

2.Late Fee Calculation

If payment is made after the due date, a penalty (late fee) is added.

Example: ₹50 per day delay.

3.Discount Handling

Applies discounts if available (e.g., no-claim bonus, long-term customer discount).

4.Tax Calculation

Adds applicable taxes (like GST).

5.Total Bill Generation

Final amount = Premium + Late Fee + Tax − Discount

6.Bill Display

Shows detailed bill:

Policy ID

Policyholder Name

Premium Amount

Due Date

Late Fee

Total Amount

4. Invoice Generation Module

The Invoice Generation Module is responsible for creating and managing invoices for policyholders based on their premium payments. It ensures accurate billing, transparency, and proper record-keeping.
Key Functions:

1.Automatic Invoice Creation

Generates invoices when a premium is due or a payment is made.

2.Policy Details Inclusion

Includes important information such as:

Policy ID

Policyholder Name

Premium Amount

Due Date

Payment Status

5. Exit Module

The Exit Module is responsible for safely terminating the application when the user chooses to close the system. This module ensures that all ongoing operations are completed properly and any necessary data is saved before exiting.

1. METHODOLOGY

The Insurance Policy Management and Premium Reminder System is developed using a structured approach to ensure efficient policy handling and timely premium notifications. The methodology followed in this system includes the following steps:

1.Storing policy details (Policy ID, holder name, premium amount, due date)

2.Managing customer information

3.Sending reminders before premium due dates

4.1. Problem Analysis

The traditional insurance management process, most operations such as policy tracking, premium payment monitoring, and customer record maintenance are handled manually or through disconnected systems. This leads to several challenges for both insurance companies and policyholders.

One major problem is the lack of proper organization of policy data. Managing large volumes of customer and policy information manually increases the risk of errors, data loss, and duplication. It becomes difficult to retrieve or update records efficiently.

Another critical issue is missed premium payments. Policyholders often forget due dates due to lack of timely reminders, which may result in policy lapses, penalties, or loss of benefits. Existing systems may not provide automated notifications, making the process unreliable.

4.2. System Design

The Insurance Policy Management and Premium Reminder System is designed to efficiently manage customer insurance policies, track premium payments, and send timely reminders. The system follows a modular and structured design to ensure easy maintenance, scalability, and usability.

4.3. Program Development

The program development of the Insurance Policy Management and Premium Reminder System involves designing and implementing a structured application to manage policy details and notify users about

premium due dates efficiently. The development process follows a step-by-step approach including requirement analysis, system design, coding, testing, and deployment.

The premium reminder feature is implemented using date comparison logic, where the system checks the current date against the due date and generates alerts for upcoming payments. Proper validation is included to ensure accurate data entry.

4.4. Testing and Validation

Testing ensures that the program works correctly and produces the expected results.

Different test cases were used to check the functionality of the system.

Example Test Cases:

Test Case	**Expected Result**
Add new policy	policy details saved successfully
Enter premium amount	Correct amount accepted
view policy details	Correct policy information displayed
Invalid input	Error handled properly

This phase ensures that the program runs without errors and produces accurate results.

4.5. Result and Documentation

The Insurance Policy Management and Premium Reminder System was successfully developed and tested. The system performs all the required operations efficiently, including:

Adding new insurance policies

Viewing and managing existing policy details

Calculating premium amounts

Generating invoices

Providing timely reminders for premium due dates

The application reduces manual work, minimizes errors, and ensures that users never miss their premium payments. The system is user-friendly and works effectively for basic insurance management needs.

4.6. Algorithm

Algorithm for the INSURANCE POLICY MANAGEMENT & PREMIUM REMINDER SYSTEM:

1.Start

2.Enter customer details (ID, Name, Policy Type)

3.Enter policy details (Policy Amount, Rate, Due Date)

4.Calculate premium

5.Store data in system

6.Check current date

7.Compare with due date

8.If due date is near → Send reminder

9.If due date passed → Send overdue alert

10.Display policy and payment details

11.Stop

4.7. Flowchart

```
┌──────────────┐
│ Start │
└──────┬───────┘
       ↓
┌──────────────────────────┐
│ Enter Customer Data│
└──────┬───────────────────┘
       ↓
┌──────────────────────────┐
│ Enter Policy Info │
└──────┬───────────────────┘
       ↓
┌──────────────────────────┐
│ Calculate Premium │
└──────┬───────────────────┘
```

```
        ↓
┌──────────────────────────┐
│ Store Data               │
└──────────────────────────┘
        ↓
┌──────────────────────────┐
│ Check Current Date       │
└──────────────────────────┘
        ↓
┌──────────────────────────┐
│ Due Date Near?           │
└──────────────────────────┘
    │Yes      │No
    ↓         ↓
┌──────────────────┐  ┌──────────────────────┐
│Send Reminder     │  │ Overdue?             │
└──────────────────┘  └──────────────────────┘
    ↓             ↓
        ┌──────────────────┐
        │ Send Alert       │
        └──────────────────┘
            ↓
      ┌──────────────────────────┐
      │ Display Details          │
      └──────────────────────────┘
          ↓
      ┌──────────┐
      │ Stop     │
      └──────────┘
```

1. SOURCE CODE

```
import java.util.*;

class Policy {
    int policyId;
    String holderName;
    double premiumAmount;
    String dueDate;

    Policy(int policyId, String holderName, double premiumAmount, String dueDate) {
        this.policyId = policyId;
        this.holderName = holderName;
        this.premiumAmount = premiumAmount;
        this.dueDate = dueDate;
    }

    void display() {
        System.out.println("Policy ID: " + policyId);
        System.out.println("Holder Name: " + holderName);
        System.out.println("Premium Amount: " + premiumAmount);
        System.out.println("Due Date: " + dueDate);
        System.out.println("--------------------------");
    }
}

public class InsuranceSystem {
    public static void main(String[] args) {
```

```
Scanner sc = new Scanner(System.in);
ArrayList<Policy> policies = new ArrayList<>();

int choice;

do{
    System.out.println("\n--- Insurance Policy System ---");
    System.out.println("1. Add Policy");
    System.out.println("2. View Policies");
    System.out.println("3. Premium Reminder");
    System.out.println("4. Exit");
    System.out.print("Enter choice: ");
    choice = sc.nextInt();

    switch (choice) {

        case 1:
            System.out.print("Enter Policy ID: ");
            int id = sc.nextInt();
            sc.nextLine();

            System.out.print("Enter Holder Name: ");
            String name = sc.nextLine();

            System.out.print("Enter Premium Amount: ");
            double amount = sc.nextDouble();
            sc.nextLine();

            System.out.print("Enter Due Date (dd-mm-yyyy): ");
            String date = sc.nextLine();
```

```
    policies.add(new Policy(id, name, amount, date));
    System.out.println("Policy Added Successfully!");
    break;

case 2:
    if(policies.isEmpty()) {
        System.out.println("No policies found.");
    } else {
        for(Policy p : policies) {
            p.display();
        }
    }
    break;

case 3:
    System.out.print("Enter today's date (dd-mm-yyyy): ");
    String today = sc.next();

    System.out.println("\n--- Premium Reminders ---");
    for(Policy p : policies) {
        if(p.dueDate.equals(today)) {
            System.out.println("Reminder: Premium due for " + p.holderName);
        }
    }
    break;

case 4:
    System.out.println("Exiting...");
    break;
```

```
            default:
                System.out.println("Invalid choice!");
        }

    }while (choice != 4);

    sc.close();
  }
}
```

1. INPUT AND OUTPUT SCREENS

```
----------------------------------------
INSURANCE POLICY INPUT SCREEN
----------------------------------------

Policy ID       : [___________]
Holder Name     : [___________]
Insurance Type : [Dropdown ▼]
Premium Amount : [___________]
Due Date        : [__/__/____]
Mobile Number : [___________]
Email ID        : [___________]

----------------------------------------
[ Add ] [ Update ] [ Delete ]
[ Clear ]          [ Exit ]
----------------------------------------
```

OUTPUT SCREENS

--- Insurance Policy System ---

1. Add Policy

2. View Policies

3. Premium Reminder

4. Exit

Enter choice: 1

Enter Policy ID: 101

Enter Holder Name: Harini

Enter Premium Amount: 5000

Enter Due Date (dd-mm-yyyy): 10-04-2026

Policy Added Successfully!

--- Insurance Policy System ---

Enter choice: 2

Policy ID: 101

Holder Name: Harini

Premium Amount: 5000.0

Due Date: 10-04-2026

--- Insurance Policy System ---

Enter choice: 3

Enter today's date (dd-mm-yyyy): 10-04-2026

--- Premium Reminders ---

Reminder: Premium due for Harini

--- Insurance Policy System ---

Enter choice: 4

Exiting...

1. DISCUSSION & CONCLUSION

7.1 Discussion

The Insurance Policy Management and Premium Reminder System is designed to simplify and automate the process of managing insurance policies and tracking premium payments. Traditionally, maintaining policy records and remembering due dates can be difficult and error-prone when done manually. This system addresses those challenges by providing a structured and digital solution.

The system allows users to store and manage important policy details such as policy ID, policyholder name, premium amount, and due date. By organizing data efficiently, it reduces the risk of losing important information and improves accessibility.

7.2 Conclusion

The Insurance Policy Management and Premium Reminder System is an efficient and user-friendly solution designed to simplify the management of insurance policies and ensure timely premium payments. This system reduces manual effort by automating key tasks such as policy storage, premium calculation, due date tracking, and reminder notifications.

By organizing all policy-related information in a structured manner, the system improves accuracy, minimizes the risk of missed payments, and enhances overall productivity. The reminder feature plays a crucial role in alerting users about upcoming due dates, thereby avoiding penalties and policy lapses.

Additionally, the system is scalable and can be further enhanced with advanced features such as mobile notifications, cloud storage, and integration with payment gateways. Overall, this project demonstrates how technology can streamline insurance management processes, making them more reliable, efficient, and accessible for both users and administrators

1. REFERENCES

The following resources were used during the development of this project.

Books Herbert Schildt – Java: The Complete Reference

E. Balagurusamy – Programming with Java

Websites

Oracle Corporation – https://www.oracle.com/java/

GeeksforGeeks – https://www.geeksforgeeks.org

W3Schools – https://www.w3schools.com

Javatpoint – https://www.javatpoint.com